

Asteroids



Starter:

<https://scratch.mit.edu/projects/1309089534/editor>

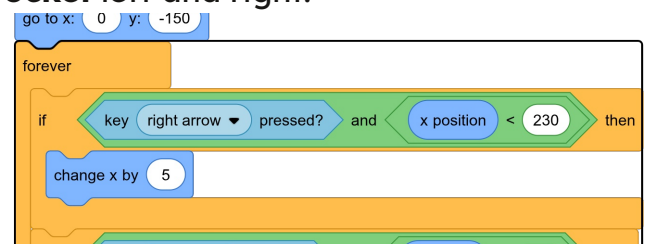
Don't forget to remix and rename.

Remix

1 Your spaceship is flying through a dangerous asteroid storm! Rocks are falling from space and you must steer your ship to avoid them. How long can you survive before you crash?

2 Use the left and right keys to move the **rocket** left and right.

- The picture on the right shows how to move the rocket right. Make sure you make it move left as well.
- Notice how it stops the rocket running too far off the side.

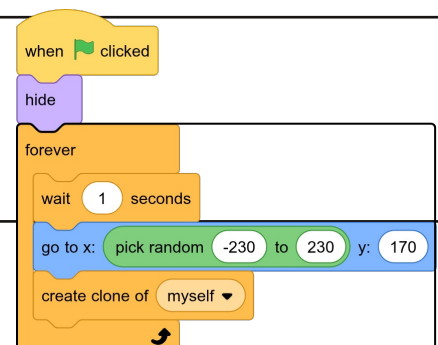


3 Now to create the asteroid storm.

- This will make a new asteroid as a clone every one second.
- The asteroids are being created, but they are still hidden - we will fix that in the next step.

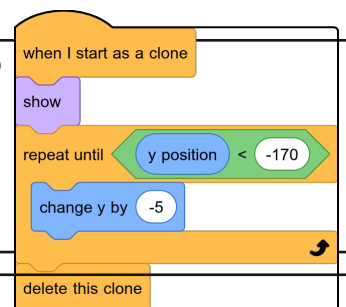
Notice

Every asteroid is a **clone** and they all run the same code. This is how we can have lots of asteroids at once.



4 The last step creates lots of new hidden asteroids at the top of the stage. Now to make them fall. This script will;

- Show the new asteroid
- Make the asteroid fall until it hits the bottom
- Delete the clone.



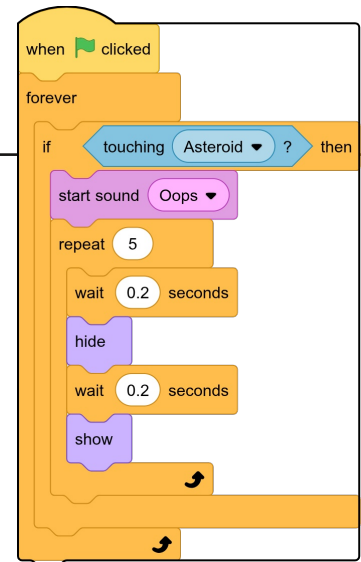
5 Some things to try;

- Can you make the new asteroids appear at random times instead of every second?
- The asteroid sprite has four costumes - can you make each new asteroid use a different costume?
- How can you make the asteroids fall at different speeds? Hint: you will need to use a variable **for this sprite only**.

6

Create a new script for the **rocket** to check when it hits an asteroid.

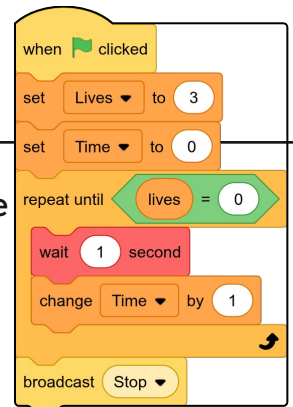
- Use a **forever** loop to keep checking.
- Detect when the asteroid is hit and play a sound.
- Flash the rocket for a couple of seconds while the player can get the rocket to safety.



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At the moment the game will go on forever. Let's add three lives and see how long the player can last. Add the script on the right to the stage;

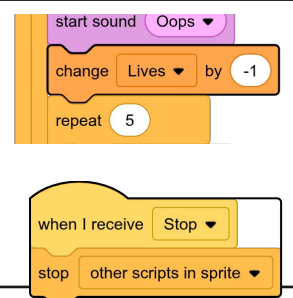
- Create a new variable **Lives** and set it to 3 at the start.
- Create a new variable **Time** and set it to 0 at the start.
- Adds 1 to the time every second.
- Only do this while **Lives** is greater than 0.
- Broadcast a message called **Stop**. All sprites should listen for this message and stop their other scripts.



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Now you need to change **Lives** when the rocket is hit and then make all the other sprites to stop what they are doing when they get the **Stop** message.

- In the rocket update the script that detects that it has been hit to take 1 off **Lives** each time.
- Add the script on the right to **all** sprites so that all their other scripts stop when they get the **Stop** message.



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What can you add to the game? Here are some ideas;

- There is a laser sprite. You could let the rocket defend itself by blasting the asteroids.
- Can you make the game get harder as you go along? Maybe the asteroids can get faster, or appear more often.
- There could be other object that the rocket can catch to do special things.
- Something else from your own imagination!

Thank you to <https://www.kenney.nl/> for the sprites used in this game.